

# Convex Sets and Convex Functions

## Learning objectives

By the end of this chapter, you should be able to

- verify convexity of a set from the line-segment definition;
- build new convex sets by intersection;
- distinguish convex, strictly convex, concave, and strictly concave functions;
- recognize convex sublevel sets and interpret convexity through epigraphs;
- state the local-to-global payoff of convex minimization;
- apply Jensen's inequality to finite convex combinations; and
- establish convexity using sums, nonnegative scaling, pointwise maxima, and composition rules.

Convexity is the geometric structure that makes many nonlinear optimization problems tractable. In a convex feasible set, movement along a straight line between feasible decisions remains feasible. For a convex objective, the value at an intermediate point is no larger than the corresponding linear interpolation of endpoint values. This geometry already points toward global optimality; later chapters develop the corresponding certificates and algorithms.

## 3.1 Convex sets

**Definition 3.1** (Convex set). A set  $C \subseteq \mathbb{R}^n$  is *convex* if

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C \quad \text{for every } \mathbf{x}, \mathbf{y} \in C \text{ and } \lambda \in [0, 1].$$

The set

$$[\mathbf{x}, \mathbf{y}] := \{\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} : \lambda \in [0, 1]\}$$

is the line segment joining  $\mathbf{x}$  and  $\mathbf{y}$ . Thus  $C$  is convex precisely when it contains the segment joining every pair of its points. See [Figure 3.1](#) for an illustration.

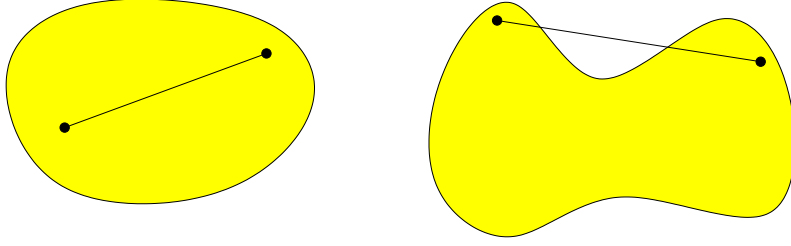


Figure 3.1: A line segment remains inside a convex set but may leave a nonconvex set.

### Warning

Convexity governs line segments; openness and closedness govern neighborhoods and limits, while boundedness controls size. No implication holds in either direction between convexity and any of these three properties. In  $\mathbb{R}$ , the set  $[0, 1)$  is convex but neither open nor closed, and  $\mathbb{R}$  is convex but unbounded. Conversely,  $\mathbb{R} \setminus \{0\}$  is open but not convex, while the unit circle

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$$

is closed and bounded but not convex: it contains  $(1, 0)$  and  $(-1, 0)$ , but not their midpoint.

**Example 3.1** (A linear-inequality set). Let  $\mathbf{a} \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ . The set

$$H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} \leq \alpha\}$$

is convex. When  $\mathbf{a} \neq \mathbf{0}$ , it is a half-space. Indeed, if  $\mathbf{x}, \mathbf{y} \in H$  and  $\lambda \in [0, 1]$ , then

$$\begin{aligned} \mathbf{a}^\top (\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) &= \lambda \mathbf{a}^\top \mathbf{x} + (1 - \lambda) \mathbf{a}^\top \mathbf{y} \\ &\leq \lambda \alpha + (1 - \lambda) \alpha = \alpha. \end{aligned}$$

**Example 3.2** (A Euclidean ball). For  $\mathbf{a} \in \mathbb{R}^n$  and  $r > 0$ , the closed ball

$$\overline{B}(\mathbf{a}, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| \leq r\}$$

is convex. If  $\mathbf{x}, \mathbf{y}$  belong to the ball and  $\lambda \in [0, 1]$ , the triangle inequality and absolute homogeneity give

$$\begin{aligned} \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} - \mathbf{a}\| &= \|\lambda(\mathbf{x} - \mathbf{a}) + (1 - \lambda)(\mathbf{y} - \mathbf{a})\| \\ &\leq \lambda \|\mathbf{x} - \mathbf{a}\| + (1 - \lambda) \|\mathbf{y} - \mathbf{a}\| \\ &\leq \lambda r + (1 - \lambda) r = r. \end{aligned}$$

The same proof, with a strict final inequality, shows that an open ball is convex.

**Example 3.3** (A set with a curved boundary). Consider

$$S := \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \geq 5, x_1^2 - 4x_2 \leq 7\}.$$

Let  $\mathbf{x}, \mathbf{y} \in S$ ,  $\lambda \in [0, 1]$ , and  $\mathbf{w} := \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ . Clearly  $w_2 \geq 5$ . Moreover, using

$$2x_1y_1 \leq x_1^2 + y_1^2,$$

$$\begin{aligned} w_1^2 - 4w_2 &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &\leq \lambda(x_1^2 - 4x_2) + (1 - \lambda)(y_1^2 - 4y_2) \\ &\leq 7. \end{aligned}$$

Hence  $\mathbf{w} \in S$ , so  $S$  is convex.

### 3.1.1 Intersections and polyhedra

**Proposition 3.1** (Intersections preserve convexity). *Let  $I$  be an index set. If  $\{C_i\}_{i \in I}$  is any family of convex subsets of  $\mathbb{R}^n$ , then*

$$C := \bigcap_{i \in I} C_i$$

*is convex (including the case in which the intersection is empty).*

*Proof.* Take  $\mathbf{x}, \mathbf{y} \in C$ . Then  $\mathbf{x}, \mathbf{y} \in C_i$  for every  $i \in I$ . Since each  $C_i$  is convex,  $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in C_i$  for every  $i$  and every  $\lambda \in [0, 1]$ . Therefore the convex combination belongs to their intersection.  $\square$

**Corollary 3.2** (Polyhedra are convex). *Let  $m$  be a positive integer. For  $A \in \mathbb{R}^{m \times n}$  and  $\mathbf{b} \in \mathbb{R}^m$ , the polyhedron*

$$P := \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$$

*is convex.*

*Proof.* Write  $\mathbf{b} := (b_1, \dots, b_m)^\top$  and write the rows of  $A$  as  $\mathbf{a}_i^\top$ , where  $\mathbf{a}_i \in \mathbb{R}^n$ . Then

$$P = \bigcap_{i=1}^m \{\mathbf{x} : \mathbf{a}_i^\top \mathbf{x} \leq b_i\},$$

an intersection of convex linear-inequality sets.  $\square$

Unions do not generally preserve convexity. For example,  $[-2, -1] \cup [1, 2]$  is a union of two convex sets but does not contain the segment joining  $-1$  and  $1$ .

## 3.2 Convex and concave functions

**Definition 3.2** (Convexity and strict convexity). Let  $C \subseteq \mathbb{R}^n$  be convex. A function  $f : C \rightarrow \mathbb{R}$  is *convex* if

$$f(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \quad (3.1)$$

for all  $\mathbf{x}, \mathbf{y} \in C$  and  $\lambda \in [0, 1]$ . It is *strictly convex* if the inequality is strict whenever  $\mathbf{x} \neq \mathbf{y}$  and  $\lambda \in (0, 1)$ .

Geometrically, every chord joining two points on the graph of a convex function lies on or above the graph (see Figure 3.2 for an illustration). Strict convexity excludes nontrivial straight segments in the graph. The domain must be convex so that every point at which the defining inequality is evaluated remains in the domain.

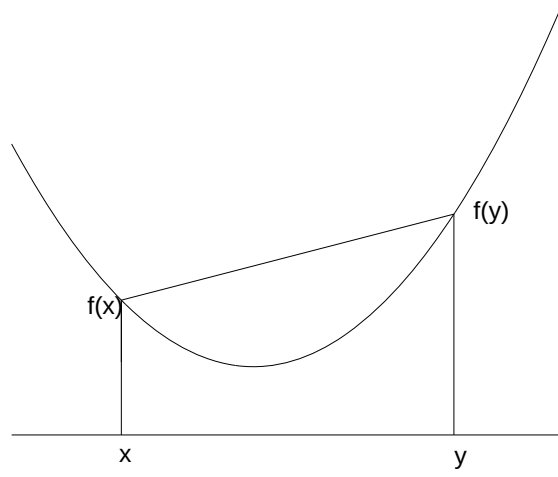


Figure 3.2: For a convex function, the graph lies below every chord joining two graph points.

**Definition 3.3** (Concavity and strict concavity). Let  $C \subseteq \mathbb{R}^n$  be convex, and let  $f : C \rightarrow \mathbb{R}$ . The function  $f$  is *concave* if, for every  $\mathbf{x}, \mathbf{y} \in C$  and  $\lambda \in [0, 1]$ ,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \geq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}),$$

and it is *strictly concave* when this inequality is strict whenever  $\mathbf{x} \neq \mathbf{y}$  and  $\lambda \in (0, 1)$ .

A function is concave if and only if its negative is convex. Thus results for convex minimization immediately have counterparts for concave maximization.

**Example 3.4** (The Euclidean norm). Define  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  by  $f(\mathbf{x}) := \|\mathbf{x}\|$ . For  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\lambda \in [0, 1]$ ,

$$\begin{aligned} f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) &= \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| \\ &\leq \lambda \|\mathbf{x}\| + (1 - \lambda) \|\mathbf{y}\|. \end{aligned}$$

It is not strictly convex as a function on  $\mathbb{R}^n$ : define the first coordinate vector by  $\mathbf{e}_1 := (1, 0, \dots, 0)^\top$ , and take  $\mathbf{x} := \mathbf{e}_1$  and  $\mathbf{y} := 2\mathbf{e}_1$ . Equality then holds for every  $\lambda \in (0, 1)$ .

**Example 3.5** (Affine functions). For  $\mathbf{a} \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ , define  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  by  $f(\mathbf{x}) := \mathbf{a}^\top \mathbf{x} + \alpha$ . For  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\lambda \in [0, 1]$ ,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) = \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Thus every affine function is both convex and concave. On a domain containing a nontrivial segment, it is neither strictly convex nor strictly concave.

Conversely, if a function on a convex set is both convex and concave, then equality holds in the chord inequality, so the function is affine on that set. On  $\mathbb{R}^n$ , it therefore has the form  $\mathbf{a}^\top \mathbf{x} + \alpha$ . It need not be linear: linearity additionally requires  $\alpha = 0$ , equivalently  $f(\mathbf{0}) = 0$ .

**Example 3.6** (A convex function that is not strict). Define  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  by  $f(x_1, x_2) := x_1^2 - 4x_2$ . For  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$  and  $\lambda \in [0, 1]$ , the same calculation used for the curved set gives

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

It is not strictly convex because equality holds for distinct points sharing the same first coordinate.

**Example 3.7** (A function that is neither convex nor concave). Define  $f : [-2, 3] \rightarrow \mathbb{R}$  by  $f(x) := (x - 1)(x - 2)(x + 1)$ . The midpoint tests

$$f(-1) = 0 > \frac{f(-2) + f(0)}{2} = -5, \quad f(1) = 0 < \frac{f(0) + f(2)}{2} = 1$$

rule out convexity and concavity, respectively.

### 3.2.1 Epigraphs and sublevel sets

**Definition 3.4** (Epigraph). Let  $D \subseteq \mathbb{R}^n$  and let  $f : D \rightarrow \mathbb{R}$ . The *epigraph* of  $f$  is

$$\text{epi } f := \{(\mathbf{x}, t) \in D \times \mathbb{R} : f(\mathbf{x}) \leq t\}.$$

Thus the epigraph consists of all points on and above the graph of  $f$ .

**Definition 3.5** (Sublevel set). Let  $D \subseteq \mathbb{R}^n$ , let  $f : D \rightarrow \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . The  $\alpha$ -*sublevel set* of  $f$  is

$$L_\alpha(f) := \{\mathbf{x} \in D : f(\mathbf{x}) \leq \alpha\}.$$

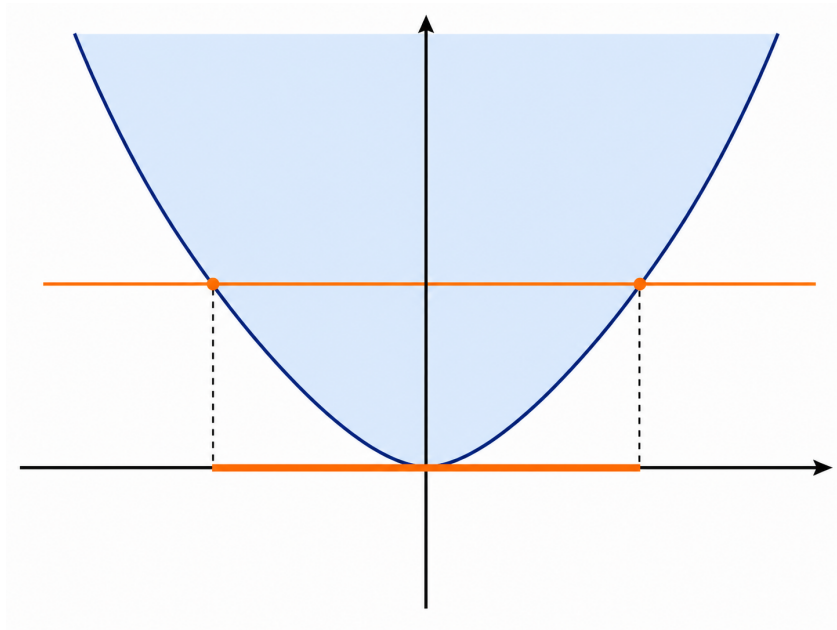


Figure 3.3: For the pictured convex function, the blue region is  $\text{epi } f$ . The orange line marks  $t = \alpha$ ; its portion inside the epigraph projects onto the orange interval  $L_\alpha(f)$ .

**Theorem 3.3** (Epigraph characterization). *Let  $D \subseteq \mathbb{R}^n$  be convex, and let  $f : D \rightarrow \mathbb{R}$ . Then  $f$  is convex if and only if  $\text{epi } f$  is a convex subset of  $\mathbb{R}^{n+1}$ .*

*Proof.* Suppose first that  $f$  is convex. Take  $(\mathbf{x}, s), (\mathbf{y}, t) \in \text{epi } f$  and  $\lambda \in [0, 1]$ . Then  $f(\mathbf{x}) \leq s$ ,  $f(\mathbf{y}) \leq t$ , and

$$\begin{aligned} f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) &\leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \\ &\leq \lambda s + (1 - \lambda)t. \end{aligned}$$

Hence the convex combination of  $(\mathbf{x}, s)$  and  $(\mathbf{y}, t)$  belongs to the epigraph.

Conversely, suppose that  $\text{epi } f$  is convex. For  $\mathbf{x}, \mathbf{y} \in D$  and  $\lambda \in [0, 1]$ , the points  $(\mathbf{x}, f(\mathbf{x}))$  and  $(\mathbf{y}, f(\mathbf{y}))$  lie in the epigraph. Their convex combination therefore lies there as well, so

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}).$$

Thus  $f$  is convex. □

**Proposition 3.4** (Convex functions have convex sublevel sets). *Let  $D \subseteq \mathbb{R}^n$  be convex, let  $f : D \rightarrow \mathbb{R}$  be convex, and let  $\alpha \in \mathbb{R}$ . Then  $L_\alpha(f)$  is convex.*

*Proof.* Take  $\mathbf{x}, \mathbf{y} \in L_\alpha(f)$  and  $\lambda \in [0, 1]$ . Convexity of  $D$  gives  $\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in D$ , while convexity of  $f$  gives

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \leq \alpha.$$

Therefore  $\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in L_\alpha(f)$ . □

Geometrically, a sublevel set is a horizontal slice of the epigraph. For a function  $f : D \rightarrow \mathbb{R}$  and  $\alpha \in \mathbb{R}$ ,

$$L_\alpha(f) \times \{\alpha\} = \text{epi } f \cap (\mathbb{R}^n \times \{\alpha\}).$$

#### Warning

The converse of [Proposition 3.4](#) is false: convexity of every sublevel set does not imply convexity of the function. For example, define  $q : \mathbb{R} \rightarrow \mathbb{R}$  by  $q(x) := x^3$ . Although  $q$  is not convex,  $L_\alpha(q) = (-\infty, \sqrt[3]{\alpha}]$  is convex for every  $\alpha \in \mathbb{R}$ . Such functions are called *quasiconvex*; quasiconvexity is weaker than convexity.

Returning to the curved set considered earlier, define  $g : \mathbb{R}^2 \rightarrow \mathbb{R}$  by  $g(x_1, x_2) := x_1^2 - 4x_2$ . Then

$$\{\mathbf{x} \in \mathbb{R}^2 : x_2 \geq 5, x_1^2 - 4x_2 \leq 7\} = \{\mathbf{x} \in \mathbb{R}^2 : x_2 \geq 5\} \cap L_7(g).$$

Its convexity follows at once from [Propositions 3.1](#) and [3.4](#).

#### Important note: the local-to-global payoff

If a convex function is minimized over a nonempty convex set, every local minimizer is global. If the function is strictly convex, a global minimizer, when it exists, is unique. The complete proof is given in [Theorem 4.1](#).

### 3.3 Finite convex combinations and Jensen's inequality

Let  $k \geq 2$  be an integer, let  $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \in \mathbb{R}^n$ , and let  $\lambda_1, \dots, \lambda_k \geq 0$  satisfy  $\sum_{j=1}^k \lambda_j = 1$ . The point

$$\mathbf{x} := \sum_{j=1}^k \lambda_j \mathbf{x}^{(j)}$$

is a *convex combination* of  $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}$ . An induction on  $k$  using the two-point definition shows that every convex combination of points in a convex set also belongs to that set.

**Theorem 3.5** (Finite Jensen inequality). *Let  $k \geq 2$  be an integer, let  $C \subseteq \mathbb{R}^n$  be convex, and let  $f : C \rightarrow \mathbb{R}$  be convex. For any  $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \in C$  and weights  $\lambda_1, \dots, \lambda_k \geq 0$  satisfying  $\sum_{j=1}^k \lambda_j = 1$ ,*

$$f\left(\sum_{j=1}^k \lambda_j \mathbf{x}^{(j)}\right) \leq \sum_{j=1}^k \lambda_j f(\mathbf{x}^{(j)}).$$

*Proof.* We argue by induction on  $k$ . The case  $k = 2$  is the definition. Suppose the result holds for  $k - 1$  points and set  $s := 1 - \lambda_k$ . If  $s = 0$ , the claim is immediate. Otherwise define

$$\hat{\mathbf{x}} := \sum_{j=1}^{k-1} \frac{\lambda_j}{s} \mathbf{x}^{(j)}.$$

The normalized coefficients are nonnegative and sum to one, so  $\hat{\mathbf{x}} \in C$ . Two-point convexity followed by the induction hypothesis yields

$$\begin{aligned} f\left(\sum_{j=1}^k \lambda_j \mathbf{x}^{(j)}\right) &= f(s\hat{\mathbf{x}} + \lambda_k \mathbf{x}^{(k)}) \\ &\leq sf(\hat{\mathbf{x}}) + \lambda_k f(\mathbf{x}^{(k)}) \\ &\leq \sum_{j=1}^{k-1} \lambda_j f(\mathbf{x}^{(j)}) + \lambda_k f(\mathbf{x}^{(k)}). \end{aligned}$$

□

Jensen's inequality is useful both analytically and conceptually: the value of a convex function at an average is no greater than the average of its values.

### 3.4 Operations that preserve convexity

**Proposition 3.6** (Sums and scalar multiples). *Let  $C \subseteq \mathbb{R}^n$  be convex, and let  $f_1, f_2 : C \rightarrow \mathbb{R}$  be convex.*

- (a)  $f_1 + f_2$  is convex.
- (b) If  $\alpha \geq 0$ , then  $\alpha f_1$  is convex.
- (c) If  $\alpha < 0$ , then  $\alpha f_1$  is concave.

*Consequently, every nonnegative weighted sum of convex functions is convex.*

*Proof.* For  $\mathbf{x}, \mathbf{y} \in C$  and  $\lambda \in [0, 1]$ , add the defining inequalities for  $f_1$  and  $f_2$  to prove (a). Multiplying the defining inequality by a nonnegative scalar proves (b); multiplication by a negative scalar reverses the inequality and proves (c).  $\square$

More generally, let  $C \subseteq \mathbb{R}^n$  be convex, let  $m$  be a positive integer, let  $f_i : C \rightarrow \mathbb{R}$  be convex for  $i = 1, \dots, m$ , and let  $\alpha_i \geq 0$  for  $i = 1, \dots, m$ . If there is an index  $j \in \{1, \dots, m\}$  for which  $f_j$  is strictly convex and  $\alpha_j > 0$ , then  $\sum_{i=1}^m \alpha_i f_i$  is strictly convex. The condition  $\alpha_j > 0$  is essential: a zero coefficient removes the strictly convex term.

**Proposition 3.7** (Pointwise maximum). *Let  $m$  be a positive integer and let  $C \subseteq \mathbb{R}^n$  be convex. If  $f_1, \dots, f_m : C \rightarrow \mathbb{R}$  are convex, then*

$$f(\mathbf{x}) := \max_{1 \leq i \leq m} f_i(\mathbf{x})$$

*is convex on  $C$ .*

*Proof.* For any  $\mathbf{x}, \mathbf{y} \in C$ ,  $\lambda \in [0, 1]$ , and every  $i$ ,

$$\begin{aligned} f_i(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) &\leq \lambda f_i(\mathbf{x}) + (1 - \lambda)f_i(\mathbf{y}) \\ &\leq \lambda \max_j f_j(\mathbf{x}) + (1 - \lambda) \max_j f_j(\mathbf{y}). \end{aligned}$$

Taking the maximum over  $i$  on the left proves the result.  $\square$

#### Warning

Pointwise maxima and pointwise minima behave differently. A pointwise minimum of convex functions need not be convex. For example, define  $f_1, f_2 : \mathbb{R} \rightarrow \mathbb{R}$  by  $f_1(x) := x$  and  $f_2(x) := -x$ . Then  $\min\{f_1(x), f_2(x)\} = -|x|$ , which is not convex.

## 3.5 Composition rules

The behavior of the outer function determines whether composition preserves curvature. The following rules make the required monotonicity explicit.

**Proposition 3.8** (Convex nondecreasing outer function). *Let  $C \subseteq \mathbb{R}^n$  be convex, let  $h : C \rightarrow \mathbb{R}$  be convex, and let  $I \subseteq \mathbb{R}$  be an interval containing  $h(C)$ . If  $g : I \rightarrow \mathbb{R}$  is convex and nondecreasing, then  $g \circ h$  is convex on  $C$ .*

*Proof.* For  $\mathbf{x}, \mathbf{y} \in C$  and  $\lambda \in [0, 1]$ ,

$$\begin{aligned} g(h(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y})) &\leq g(\lambda h(\mathbf{x}) + (1 - \lambda)h(\mathbf{y})) \\ &\leq \lambda g(h(\mathbf{x})) + (1 - \lambda)g(h(\mathbf{y})). \end{aligned}$$

The first inequality uses convexity of  $h$  and monotonicity of  $g$ ; the second uses convexity of  $g$ .  $\square$

### Warning

Convexity of both component functions is not sufficient for convexity of their composition: the monotonicity hypothesis cannot be omitted. Define  $g, h : \mathbb{R} \rightarrow \mathbb{R}$  by  $g(t) := -t$  and  $h(x) := x^2$ . The function  $g$  is affine and therefore convex, and  $h$  is convex, but  $(g \circ h)(x) = -x^2$  is not convex.

**Proposition 3.9** (Concave nonincreasing outer function). *Let  $C \subseteq \mathbb{R}^n$  be convex, let  $h : C \rightarrow \mathbb{R}$  be convex, and let  $I \subseteq \mathbb{R}$  be an interval containing  $h(C)$ . If  $g : I \rightarrow \mathbb{R}$  is concave and nonincreasing, then  $g \circ h$  is concave on  $C$ .*

*Proof.* For  $\mathbf{x}, \mathbf{y} \in C$  and  $\lambda \in [0, 1]$ , convexity of  $h$  and the fact that  $g$  is nonincreasing give

$$g(h(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y})) \geq g(\lambda h(\mathbf{x}) + (1 - \lambda) h(\mathbf{y})).$$

Concavity of  $g$  bounds the right-hand side below by  $\lambda g(h(\mathbf{x})) + (1 - \lambda)g(h(\mathbf{y}))$ .  $\square$

**Example 3.8** (Two compositions). Define  $h : \mathbb{R} \rightarrow \mathbb{R}$  by  $h(x) := x^2 + x$ ; this function is convex on  $\mathbb{R}$ .

- (i) Define  $g_1 : \mathbb{R} \rightarrow \mathbb{R}$  by  $g_1(t) := e^t$ . Since  $g_1$  is convex and increasing,  $g_1 \circ h = e^{x^2+x}$  is convex on  $\mathbb{R}$ .
- (ii) On  $[0, \infty)$ , we have  $h(x) \geq 0$ . Define  $g_2 : [0, \infty) \rightarrow \mathbb{R}$  by  $g_2(t) := t^2$ . Since  $g_2$  is convex and nondecreasing,  $g_2 \circ h = (x^2 + x)^2$  is convex on  $[0, \infty)$ .

### Chapter summary

- A convex set contains every line segment joining two of its points.
- Intersections preserve convexity; unions generally do not. Half-spaces, balls, and polyhedra are convex.
- A convex function lies below its chords. Negating a convex function produces a concave function.
- A function is convex exactly when its epigraph is convex. Every sublevel set of a convex function is convex, but the converse fails.
- In convex minimization, every local minimizer is global.
- Jensen's inequality extends the two-point definition to finite convex combinations.
- Nonnegative sums and pointwise maxima preserve convexity.
- For compositions, curvature alone is insufficient: the required monotonicity of the outer function must also be checked.

### 3.6 Exercises

**Exercise 3.1** (Unions). State a condition under which the union of two convex sets is convex.

**Exercise 3.2** (Polyhedra by definition). Let  $m$  be a positive integer, let  $A \in \mathbb{R}^{m \times n}$ , and let  $b \in \mathbb{R}^m$ . Prove directly, without invoking [Proposition 3.1](#), that  $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq b\}$  is convex.

**Exercise 3.3** (Ball intersected with a half-space). Show that

$$\{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| \leq 5, x_1 + 2x_2 - x_3 \leq 3\}$$

is convex.

**Exercise 3.4** (Outside a ball). Let  $\mathbf{a} \in \mathbb{R}^n$  and  $r > 0$ . Is

$$\{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| \geq r\}$$

convex? Justify your answer.

**Exercise 3.5** (Graphing the Euclidean norm). For  $n = 2$ , sketch the graph of  $f(\mathbf{x}) := \|\mathbf{x}\|$  and several of its level sets. Relate their geometry to the equality case exhibited in the Euclidean-norm example.

**Exercise 3.6** (Epigraph of an affine function). Define  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  by  $f(x_1, x_2) := x_1 - 2x_2$ . Write  $\text{epi } f$  explicitly as a half-space in  $\mathbb{R}^3$ , identify its boundary plane, and verify its convexity directly.

**Exercise 3.7** (A paraboloid and a saddle). Define  $q, h : \mathbb{R}^2 \rightarrow \mathbb{R}$  by

$$q(x_1, x_2) := x_1^2 + x_2^2, \quad h(x_1, x_2) := x_1^2 - x_2^2.$$

Sketch representative sublevel sets and the epigraph of each function. Relate the geometry to the convexity of  $q$  and the failure of convexity of  $h$ .